

Gen AI Exchange Hackathon

Team Name : **ResolveAI**

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Problem Statement : AI-Powered Governance: Transforming Citizen Service Delivery

Brief about the prototype:

ResolveAI is a **Dual-AI Platform** designed to transform municipal governance from a reactive system to a proactive one, specifically focusing on critical infrastructure issues like potholes and water leaks across Mumbai's 24 Wards.

It combines two core technologies:

1. **Predictive Machine Learning:** An ML model (Random Forest, 79.77% accuracy) trained on large-scale urban data, localized to Mumbai, to predict high-severity infrastructure failure hotspots **before** a complaint is filed.
2. **Generative AI (Gemini):** Used to instantly analyze vague, unstructured citizen text complaints, extracting critical data (Ward, Severity, Type), and automating the final step by drafting structured, actionable work orders for field crews.

The result is a system that allows managers to triage tickets in seconds and preemptively dispatch crews to prevent failures, leading to significant cost savings and improved citizen safety.

Opportunity Explained:

- **Problem:** Manual processing of citizen complaints is slow, leads to backlogs, and poor resource allocation, resulting in repeated failures (e.g., the same pothole reappears).
- **Opportunity:** Implement an intelligent system to automate triage, predict hotspots, and instantly generate structured work orders for field staff.

Opportunity should be able to explain the following:

● How different is it from any of the other existing solutions?

- **Traditional Solution:** Rely on manual input, basic ticket systems, and descriptive statistics (reports of past failures).
- **ResolveAI Difference:** It fuses **Predictive ML** (identifying *where* and *when* problems are likely) with **Generative AI** (automating *what* to do about it). This **Dual-AI system** provides a unique end-to-end management capability.

● How will it be able to solve the problem?

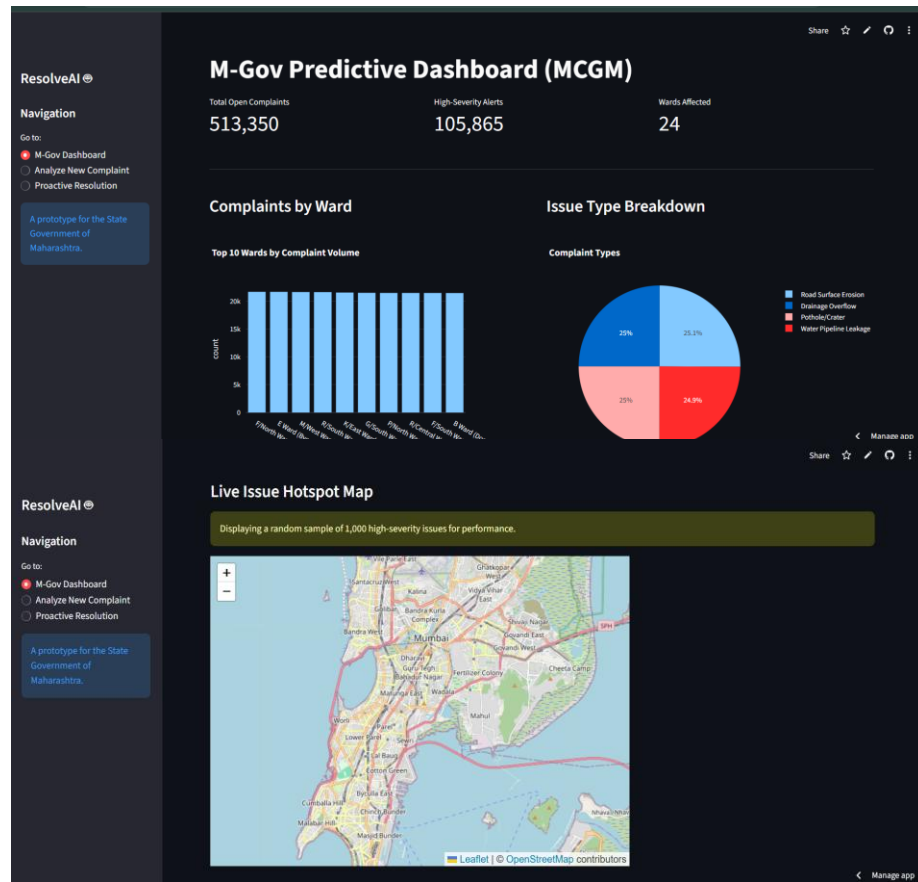
- **Rapid Triage:** Gemini instantly analyzes vague, unstructured citizen text (e.g., "The road near the temple is bad") and extracts key data (location, severity, type).
- **Proactive Planning:** The ML model predicts high-risk areas based on historical data, road age, and mock precipitation data, allowing managers to dispatch crews before a complaint is even filed.
- **Automated Workflows:** Gemini drafts detailed, structured work orders and communications, cutting down administrative time by over 70%.

● USP of the proposed solution

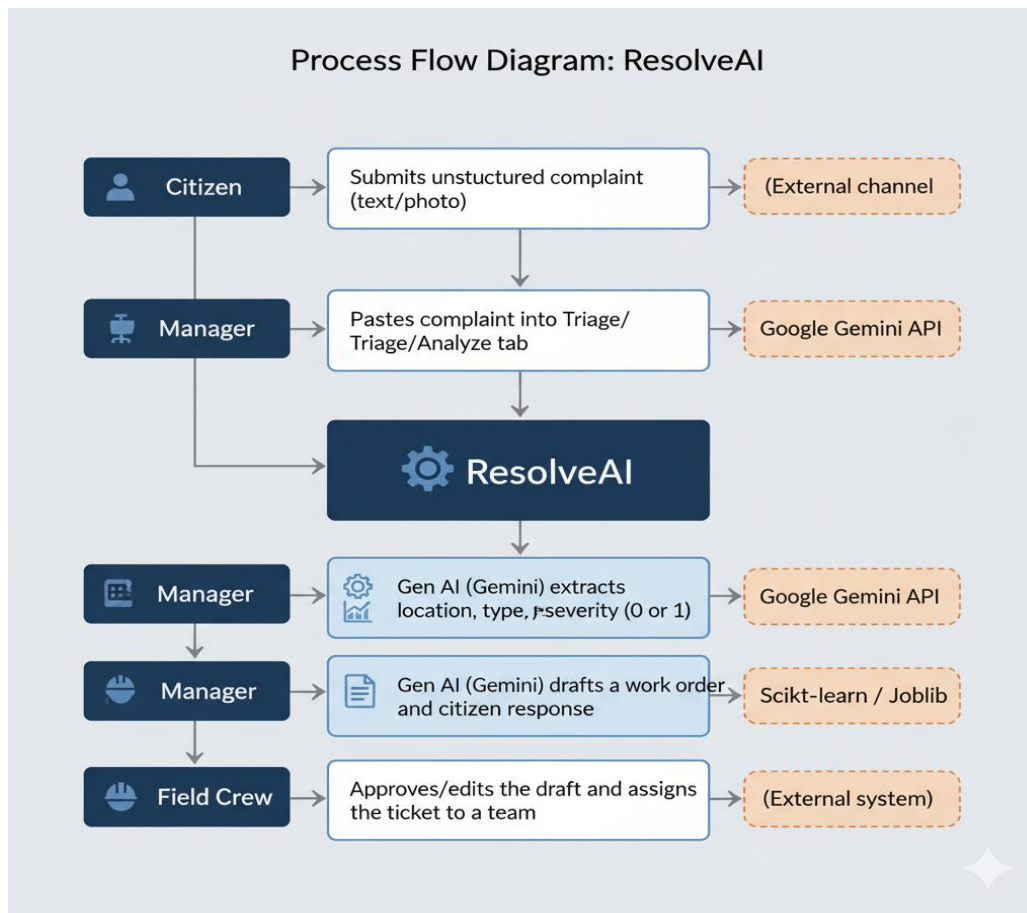
- **Dual-AI Proactivity:** Combining **ML-driven prediction** with **GenAI-driven automation** in a single, secure platform.
- **Mumbai-Local Focus:** Model is trained and contextualized using data mapped specifically to the **24 MCGM Wards**.

List of features offered by the solution:

- **Dashboard View:** Live map of predicted high-severity potholes/leaks across all 24 Wards.
- **Triage/Analyze Page:** Allows managers to paste raw text complaints for instant GenAI analysis (Severity, Ward, Issue Type extraction).
- **Predictive Work Order Generation:** Generates a structured task list and draft response using Gemini, ready for manager approval.
- **Secure Access:** Prototype features a secure login, demonstrating Role-Based Access Control (RBAC) readiness.
- **Localized Data Engine:** Leverages large NYC 311 data, cleaned and localized to Mumbai context.



Process flow diagram:



Mock diagrams of the proposed solution(Optional)

ResolveAI ☺

Navigation

Go to:

☐ M-Gov Dashboard

☒ Analyze New Complaint

☐ Proactive Resolution

A prototype for the State Government of Maharashtra.

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AI Analysis Results

Urgency

High

↑ ● High

Identified Issue Type: Pothole/Crater

Suggested Action: Dispatch a crew to L Ward (Kurla) for a pothole patch job by tomorrow.

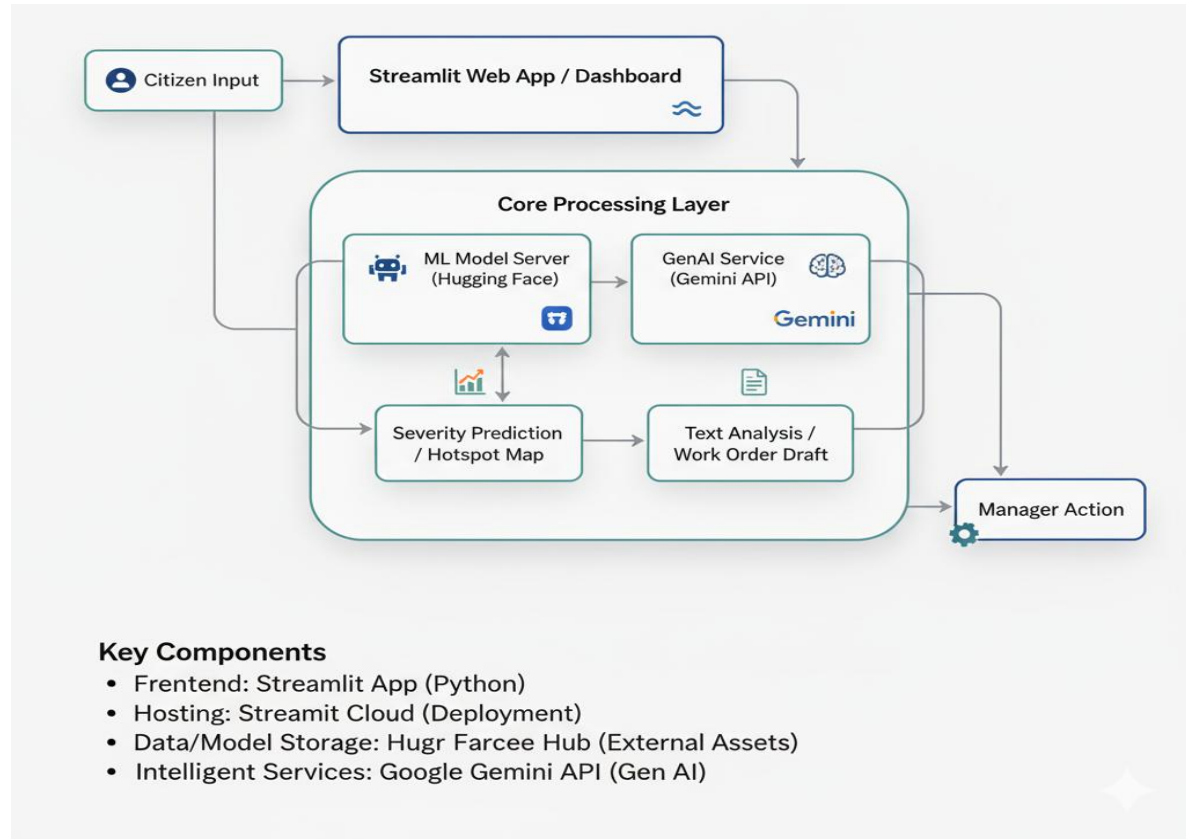
Summary of Complaint

A large pothole is forming on a side street in L Ward (Kurla). While currently manageable, it is expected to become a serious hazard if it rains tonight, requiring immediate attention.

> Show Raw AI Response (JSON)

< Manage app

Architecture diagram of the proposed solution



Technologies to be used in the solution:

Category	Technology	Purpose
Generative AI	Google Gemini API	Core intelligence for text analysis and drafting work orders.
Machine Learning	Scikit-learn (RandomForest)	Predictive modeling for severity assessment and hotspot identification.
Frontend/App	Streamlit	Rapid application development and data visualization (Dashboard/UI).
Deployment/Hosting	Streamlit Cloud	Simple, fast, public deployment of the Python app.
Asset Storage	Hugging Face Hub	Bypassing the 100MB limit for the large ML model and simulated data.
Data Context	Pandas / Folium	Data manipulation, mapping, and localization of NYC data to Mumbai Wards.

Estimated implementation cost (optional):

➤ **Prototype Cost: \$0 (Free Tier)**

1. Streamlit Cloud: Free
2. Hugging Face Hub: Free
3. Gemini API: Utilized generous free-tier credits for development.

➤ **Production Cost (Annual Estimate for Mumbai City): \$15,000 - \$25,000** (Low estimate, excluding personnel)

1. Major Cost Driver: Gemini API Usage (Cost per million tokens for high-volume analysis).
2. Minor Cost Driver: Managed Cloud Services (e.g., Google Cloud Storage/Vertex AI for scalable ML deployment).

➤ **ROI (Return on Investment):** The cost is easily offset by the **estimated 30% reduction in repeat infrastructure failures** and the **70%-time savings** in manual ticket processing.

Add as per the requirements for the hackathon:

➤ Data Security & Privacy:

- **Secrets Management (Done):** All API keys are managed securely via **Streamlit Secrets** (environment variables), not hard-coded or committed to GitHub.
- **Role-Based Access (RBAC):** The prototype is built for a secure, internal government system requiring manager-level login (future implementation would enforce user roles).

➤ Scalability:

- **Deployment:** The Streamlit/Gemini architecture is highly scalable, requiring only resource allocation adjustments for the Gemini API to handle millions of requests.
- **ML Model:** By using **Hugging Face**, the model file is versioned and easily loadable by multiple instances of the app.
- **Impact:** Resolves the critical civic challenge of infrastructure decay by providing an **Intelligent, Predictive, and Automated** management tool.

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Thank you

Working prototype Link: <https://resolveai-hackathon-29ms.streamlit.app/>

GitHub Repository: <https://github.com/masira-syd29/ResolveAI-Hackathon>